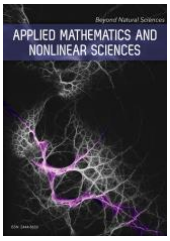
sciendo Applied Mathematics and Nonlinear Sciences, 9(1) (2024) 1-19



Applied Mathematics and Nonlinear Sciences

https://www.sciendo.com

Cluster optimization and algorithm design for machine vision in industrial robot control systems

Linyang Guo[1, 2], Runxian Yang[1, 2,†], Mingjun Tang[1], Xiaoyan Ma[2], Lixia Wang[3]

1. College of Intelligent Manufacturing, Yangzhou Polytechnic Institute, Yangzhou, Jiangsu, 225000, China.
2. Jiangsu Province Engineering Research Center of Intelligient Application for Advanced Plastic Forming, Yangzhou, Jiangsu, 225000, China.
3. Jiangsu Shuguang Electro-Optics Co., LTD., Yangzhou Jiangsu, 225000, China.

Submission Info

Communicated by Z. Sabir
Received March 30, 2024
Accepted July 17, 2024
Available online September 3, 2024

Abstract

Machine vision technology improves the ability to detect the environment of industrial robots, which contributes to the improvement of the collaboration efficiency of swarm robots. This paper proposes a roadmap extraction algorithm that utilizes the improved Mean Shift algorithm to extract the roadmap information from images acquired by binocular cameras. Subsequently, the IEO algorithm with K-Means++ is used to optimize the task allocation of the swarm robots. The experiments show that the average error of this paper's algorithm's road sign extraction is 0.025m, the ratio of full scene and homing reaches 90.6%, and the results of the scheduling algorithm under the three kinds of task volume are 59.89, 773.08, and 2704.67. The efficiency of scheduling task completion in dispensing experiments is 9.56% higher than that of the comparative algorithms. The experiment proves that the algorithm proposed in this paper has good performance and practical effects on optimizing the industrial robot control system.

Keywords: Machine vision; Waypoint extraction; Mean Shift algorithm; IEO algorithm; Robot cluster optimization.
AMS 2010 codes: 08A02

†Corresponding author.
Email address: chirs0232@sina.com

ISSN 2444-8656
sciendo
https://doi.org/10.2478/amns-2024-2539

OPEN ACCESS
© 2024 Linyang Guo, Runxian Yang, Mingjun Tang, Xiaoyan Ma and Lixia Wang, published by Sciendo.
This work is licensed under the Creative Commons Attribution alone 4.0 License.

1 Introduction

In today's increasingly competitive globalized manufacturing industry, the primary goal of the manufacturing industry is to enhance its competitiveness by using cutting-edge information and communication technologies and artificial intelligence technologies. Currently, many countries have proposed smart manufacturing strategies that aim to promote advanced manufacturing through smart technologies such as the Internet, cloud computing, big data, and the Internet of Things [1-3].

Industrial robots are an important component of FMS (flexible manufacturing system). Robots have good adaptability and flexibility and are an indispensable and important part of the automated production system. As the level of automated production and manufacturing continues to improve, the requirements for production line flexibility are also increasingly high, and the requirements for expanding the application areas of industrial robots and improving their flexibility and autonomy are also increasingly urgent [4-5]. To improve the robot's "intelligence" so that it has a certain degree of autonomy, the first thing to do is to improve the robot's understanding of its environment, so people began to install sensors for the robot to improve the robot's ability to perceive the environment. From the world's decades of research and development examples of intelligent robots, it is easy to see that the most important of the robot's sensory ability is the robot's vision. The eye is an important organ of human perception, and vision is an important means of human observation and cognition of the world. According to statistics, the total information we obtain from the surrounding environment through vision accounts for about 80% of the total information, which indicates both the large amount of information obtained by vision and the importance of the visual function to human production and life [6-8].

Robot vision is to mimic the human eye and apply machine vision technology to robots. There are reasons why machine vision technology can be successfully combined with robotics. Machine vision can quickly obtain information about the surrounding environment, but also than other sensors to get a large amount of information. Automated processing of information is convenient and easy to integrate with the robot's control system. Machine vision can quickly detect the product, especially when detecting high-speed movement of objects, such as the product line. Machine vision can greatly improve the detection efficiency. In terms of accuracy, machine vision has a clear advantage because it can reach one-thousandth of an inch accuracy, which is also greatly superior to the robots. It can greatly improve the detection efficiency. In terms of accuracy, machine vision has a clear advantage because its accuracy can reach one-thousandth of an inch, which is also greatly superior to other detection methods [9-11]. Because of the above characteristics, robots with vision will be able to work more efficiently and stably in automated production lines, and gradually into various industries, with a broader space for development.

Industrial experts have predicted that in the next 20 to 50 years, machine vision will reach a rapid development stage and become a universal technology across all industries. Machine vision systems can be used to detect almost all the products produced [12-13]. Therefore, the research and development of vision robotics technology is of great practical significance for people's lives.

After more than 50 years of development, robotics technology in the industrial field has been widely used and has become an important symbol of industrial modernization. At this stage, most industrial robots still utilize the "teaching-reproduction" method to complete the work; the robot can only complete the point-to-point action, and the flexibility of the production line is poor. Machine vision has a large amount of information, high detection accuracy, wide detection range, and other characteristics. The application of machine vision to industrial robots can enable the robot to autonomously complete the detection and judgment of the target and improve the robot's ability to adapt to the external environment [14]. Literature [15] elaborated that artificial intelligence

technology combined with computer vision can effectively assist human task execution and management decision-making. Literature [16] reveals the role played by machine vision technology in the quality control of gear industrial manufacturing and develops a new machine vision application program (Vision2D), which, after numerical detection in simulation experiments, improves the accuracy of identification and detection of gear quality in the industrial production process. Literature [17] introduces a strategy for automatic material identification based on machine vision technology, which effectively assists industrial robots in material classification and sorting, and validation experiments were carried out to evaluate the introduced strategy, pointing out that the vision identification scheme achieves the purpose of accurate identification and sorting of materials. Literature [18] explores the performance of camera stereo vision, analyzes in depth the algorithms in the machine vision chain as well as the overall profile, and uncovers the advantages and drawbacks of envisioned stereo vision in performing complex machine vision tasks. Literature [19] proposes the use of machine vision models to realize the monitoring of product quality in industrial production and sorting defective products for reprocessing to meet the requirements of high-quality production in the 4.0 industry, and then systematically evaluates the production model, and the evaluation results affirm the scientific and reasonable ideas of the industrial production model.

Industrial robots are important equipment in industrial production, with the advantages of compact structure, high flexibility, and good programmability. However, at present, most of the mainstream industrial robot control system adopts the embedded control scheme, which is highly coupled with the hardware platform and software sy. The interface standards, operation details, and underlying algorithms of each system are different. The closure of the industrial robot control system brought about by commercial competition and technical details has hindered the development of the intelligentization of industrial equipment. Literature [20] talks about the intelligent control of industrial robots becoming the trend of future development and tries to explore the ecological position of industrial robots in the industrial production chain from the perspectives of industrial robots' production goals and production demands. Literature [21], in the report on the practice of industrial robot technology, introduces the practical performance of industrial robots in industrial production, as well as the technical fields involved in industrial robots, and finally gives an outlook on the trend of research and application of industrial robots, which deepens the understanding of the practical application of industrial robots and research paths. Literature [22] designed a distributed control framework with dynamic reorganization of the production process, which can take into account the hardware control services and advanced control services, realize the automatic construction of individual tasks, and have a high degree of scalability in the practice of human-robot collaboration. Literature [23] tries to study the intelligent predictive control algorithm for industrial processes in order to improve the control effect of industrial production and envisions a new intelligent predictive control strategy, which effectively improves the control accuracy and control effect of industrial system production. Literature [24] analyzed and optimized the modular wheels, control system, and vision sensor part of the factory material transporter, respectively, in order to realize the functions of transporter smoothness, automatic cruise, as well as positioning, and took the experimental method to verify the feasibility.

In this paper, SURF is used to extract feature points from the image notation acquired by the left and right eye cameras of the robot. The inspection principle is utilized to obtain 3D information on the spatial point set, and the spatial point coordinates are converted to global coordinates during storage. To reduce computational complexity and improve matching speed, after filtering to obtain the correct feature points, the improved Mean Shift algorithm is used to cluster the appropriate feature points into TDNL signposts. A collaborative task assignment algorithm for swarm robots is proposed in this paper. Multiple MTSP problems are clustered into a small number of independent TSP problems using K-Means++, followed by solving the shortest path of each TSP problem using the IEO algorithm, and

finally merging the optimal solutions of each TSP in order to find the optimal solution of the MTSP model. In this paper, the effect of the proposed algorithm is experimentally analyzed through several aspects, such as the accuracy of roadmap extraction, the homing effect of clustered robots, and the stability and efficiency of the task allocation algorithm.

2 Binocular image feature point extraction and matching

2.1 Algorithm for obtaining 3D information of spatial points corresponding to feature points

Because the SURF algorithm is more efficient in feature point extraction, this algorithm is used in this chapter to extract feature points from the images obtained by the left and right eye cameras separately [25]. The spatial point corresponding to the feature point pair f_{lr} is denoted as S_f, and the set consisting of all spatial points is denoted as *S*. The three-dimensional information of $S_f \in S$ can be obtained using the parallax principle [26]. Figure 1 shows the principle diagram of binocular ranging. The origin of the coordinate system centered on the left and right eyes is located at their respective optical centers, and the parameters of the left and right eyes are made the same to simplify the calculation. The optical center of the two eyes is in the same plane, and the imaging plane originally behind the optical center is placed in front of the binocular optical center at a distance of *f*. *f* is the focal length, and the corresponding images acquired by the left eye and the right eye are also established as a coordinate system. The origin is the intersection of the binocular optical axis with the plane at points O_1 and O_2, then the position of the spatial point $X(x_c, y_c, z_c)$ and the corresponding points in the left and right eye images are $X_1(u_1, v_1)$ and $X_2(u_2, v_2)$. It can be seen that the *y*-coordinates of $X_1(u_1, v_1)$ and $X_2(u_2, v_2)$ are the same, i.e., $v_1 = v_2$.

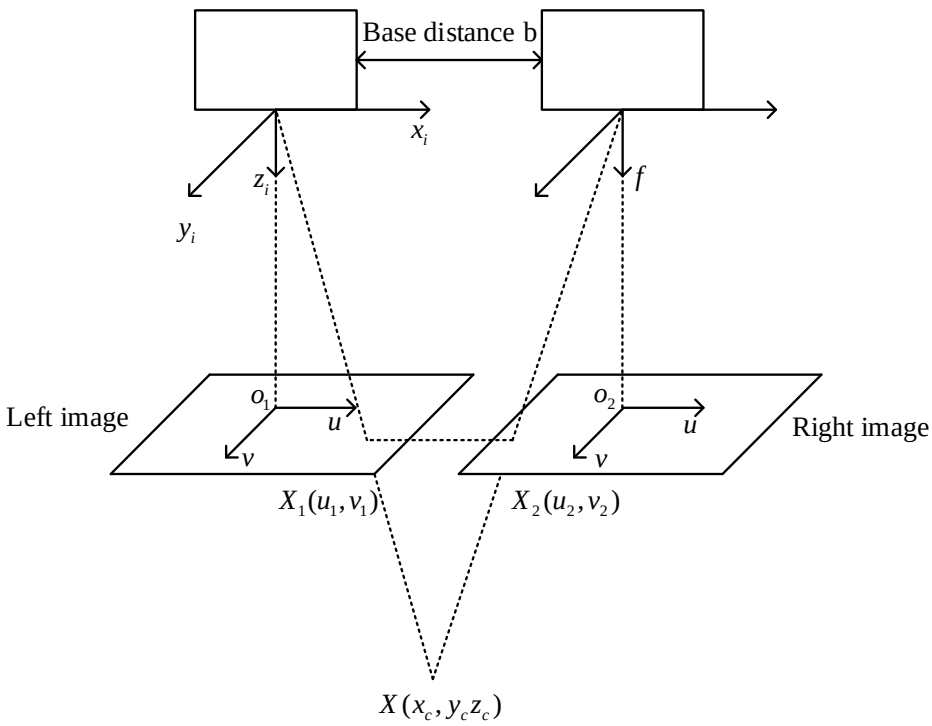


Figure 1. Basic principle of binocular vision

From the geometric relationship:

$$\begin{cases} u_1 = & f\dfrac{x_c}{z_c} \\ u_2 = & f\dfrac{x_c - b}{z_c} \\ v_1 = & v_2 = f\dfrac{y_c}{z_c} \end{cases} \tag{1}$$

Parallax can be calculated using the following formula:

$$d = u_1 - u_2 = \frac{f \cdot b}{z_c} \tag{2}$$

The coordinates of the spatial point *X* in the left-eye coordinate system are thus obtained:

$$\begin{cases} x_c = & \dfrac{b \cdot u_1}{d} \\ y_c = & \dfrac{b \cdot v_1}{d} \\ z_c = & \dfrac{b \cdot f}{d} \end{cases} \tag{3}$$

So, it is only necessary to obtain the corresponding point of the spatial point on the binocular imaging plane. At the same time, based on the internal and external parameters of the binoculars, the spatial coordinates of the point can be obtained according to the geometric relationship. This coordinate information will correspond to the feature points one by one, which will be used later in the point clustering.

After extracting the waypoints for storage, the coordinates of all spatial points in the left camera coordinate system should be converted to global coordinates, and the relationship between the two can be determined using translation vectors and rotation matrices. The robot moves in the environment generally only involves the rotation around the z-axis, so the rotation matrix can be simplified as follows: so that the chi-square coordinate of the space point in the global coordinate system is $X = (x, y, z, 1)^T$, and the chi-square coordinate in the left camera coordinate system is $X_c = (x_c, y_c, z_c, 1)^T$, and the relationship between the two can be expressed by the following equation:

$$X = (X_c^T T_1 T_2 R)^T \tag{4}$$

Where T_1 - swaps the *y* and *z* coordinates of points in the left camera coordinate system to align them with the global coordinates:

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

T_2 --The translation matrix:

$$T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\tilde{x}_c & -\tilde{y}_c & -\tilde{z}_c & 1 \end{bmatrix} \tag{6}$$

$\tilde{C}$ -- Global coordinates of the origin of the left camera coordinate system:

$$\tilde{C} = (\tilde{x}_c, \tilde{y}_c, \tilde{z}_c) \tag{7}$$

R-- a rotation matrix that rotates the spatial point about the *z*-axis by the angle β of the two coordinate systems:

$$R = \begin{bmatrix} cos\beta & -sin\beta & 0 & 0 \\ sin\beta & cos\beta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

2.2 Road sign extraction algorithm based on improved Mean Shift algorithm

After obtaining the correct feature point pairs through selection and filtering, the 3D information of the corresponding spatial points of the feature points is obtained based on binocular vision. This section then deals with the process of extracting TDNL waymarks based on this information.

Instead of directly using all qualified feature points as signposts, suitable feature points are clustered and then used as signposts, which can reduce the computational complexity during matching and make the matching speed much higher. The roadmap formed by clustering can correspond to specific objects in the environment with a large probability and can be quickly matched when the next scene appears on without repeated storage.

In this section, the 3D information of feature points is utilized to divide the feature points that are close in spatial distance into the same TDNL waypoints as much as possible so that the extracted waypoints correspond to specific objects in the environment with a large probability. This has the advantage of extracting the same waymarks with a large probability, even if the viewpoints and distances of the robots relative to the objects are different. The TDNL waymarks are essentially a set of clusters of points that are spatially close in distance.

After obtaining the coordinates of all the points in the spatial point set S in the left-eye coordinate system, a cluster analysis is performed based on the relative spatial positions of these points to extract the waymarks. In order to achieve this purpose, a suitable clustering algorithm needs to be selected.

In this paper, we use the improved Mean Shift algorithm for TDNL waypoint extraction. The Mean Shift algorithm is a kernel density estimation process. The spatial point $S_f \in S$ is regarded as the data point obtained from sampling, the number of spatial points in unit volume is regarded as the probability density, and the clustering is completed by finding the set of points with the largest probability density [27]. The probability density of each point *x* can be expressed as:

$$P(X) = \frac{1}{n}\sum_{i=1}^{n} K(X - X_i) \tag{9}$$

In order to find the point with the largest probability density, one can first derive $P(X)$ to obtain its extreme point, such that $g(x) = -k'(x)$, has:

$$P'(X) = \frac{c}{n}\sum_{i=1}^{n} k_{i'} = \frac{c}{n}\left[\sum_{i=1}^{n} g_i\right]\cdot\left[\frac{\sum_{i=1}^{n} X_i g_i}{\sum_{i=1}^{n} g_i} - X\right] \tag{10}$$

When $P'(X) = 0$, $\frac{c}{n}[\sum_{i=1}^{n} g_i] \neq 0$, order:

$$m(X) = \frac{\sum_{i=1}^{n} X_i g_i}{\sum_{i=1}^{n} g_i} - X \tag{11}$$

When $m(X)$ satisfies $m(X) \leq M$, the extreme point of the point probability density can be approximated and centered at that point. Points in a region of radius *r* form a cluster, where *M* is the threshold. If $m(X)$ of the current point *X* is larger, it means that *X* is not an extreme point, and the center point *X* should be moved to $X + m(X)$, and then the derivation should be carried out until $m(X)$ satisfies $m(X) \leq M$, which completes a clustering.

In general the clustering radius length has not grown to the maximum radius $r_{\max}$, the number of feature points n_f and $m(X)$ have met the requirements. At this time, the clusters have been formed, but there is a radius and then the appropriate expansion of the situation may still be characterized by points, and for this reason, the proposed clustering of a small expansion of the practice. If there is a point closer to the cluster and the distance from the cluster is less than $r + \delta r_e$, the radius of the cluster will be increased by δr_e to include the point, and repeat the process. The magnitude of expansion $\delta r_e \leq \delta r$ is generally taken as $\delta r_e = \frac{\delta r}{2}$. The expansion process stops if there are no more points within the next expanded cluster or if the maximum cluster radius limit $r_{\max}$ is exceeded.

The radius of the distribution range of the feature points that make up the TDNL signposts $r_{\max}$ should also be controlled within a certain range, which is consistent with the phenomenon that there

is generally a certain size limitation as an object in the natural environment, which is taken in this paper:

$$r_{max} = r_{original} + 3 \times \delta r \tag{12}$$

Style:

$r_{original}$ -- initial radius of clustering.

δr -- Radius growth step.

After completing a clustering to get a signpost, a point is randomly selected as the center of the circle from the set of feature points that have not yet been visited, and the search for the next clustering continues until the number of unvisited points $|S_{uv}| \le N_f^{\min}$.

3 Cluster robot tasking algorithm

In this subsection, a collaborative task assignment algorithm for swarm robots based on the K-Means++ algorithm and the IEO algorithm is proposed to solve the MRTA problem. First, the method uses the K-Means++ algorithm to divide the *N* task points into *K* classes, i.e., the MTSP problem is transformed into *K*-independent TSP problems. Then, the method uses the IEO algorithm to solve each TSP problem after the shortest path by merging the optimal solution of each TSP in order to find the optimal solution of the MTSP model. Figure 2 shows the flow of the method for solving the MRTA problem. The K-Means++ algorithm and the IEO algorithm utilized by the collaborative task assignment algorithm are detailed next, respectively.

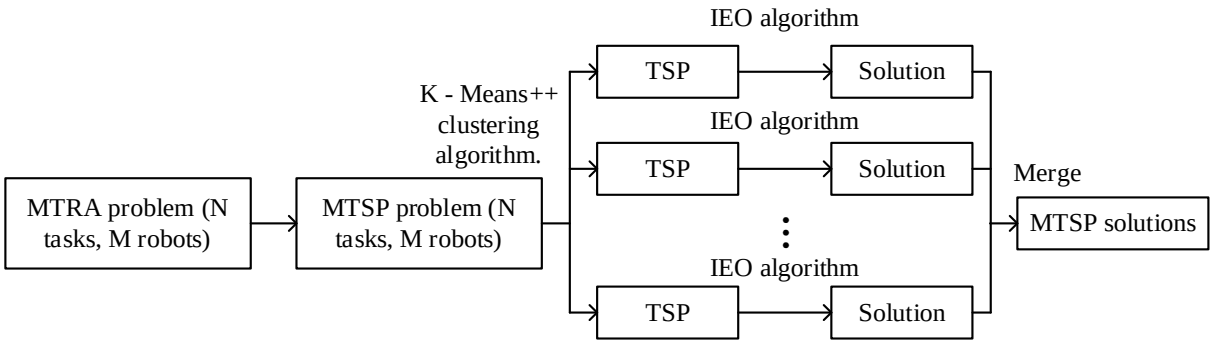


Figure 2. Collaborative task assignment policy

3.1 K-Means++ algorithm

The K-Means++ algorithm is an improved clustering algorithm for the sometimes weak clustering of the K-Means algorithm, which improves on the initial selection of cluster centers in the K-Means algorithm [28].

Since the K-Means algorithm randomly selects *K* data from the sample set as the initial clustering center may lead to the problem of weaker clustering in this algorithm, the K-Means++ algorithm has made improvements to address this problem. First, the K-Means++ algorithm randomly selects one task point in the sample set of task points as the clustering center ρ, and then selects the remaining clustering centers by repeating the following steps.

1) Calculate the distance between the remaining task points and the clustering centers that have been initialized and select the shortest of these distances, denoted as d_n.

2) Select a new task point as the new clustering center, and the principle of selection is the larger d_n is, the higher the probability that the task point is selected as the clustering center.

Repeat the above process until all K clustering centers are identified and the subsequent process is consistent with the K-Means algorithm.

3.2 IEO Algorithm

The EO algorithm is inspired by the physics of the mass balance equation, which describes the law of conservation of concentration of nonreactive substances in a control volume [29]. In this section, the EO algorithm is improved by introducing Circle chaotic mapping, Levy flight, and iterative cosine operator (ICO). The specific improvement process is described next.

1) Circle chaotic mapping. Chaos has the properties of randomness, ergodicity, and regularity. Introducing chaotic sequences can maximize the solution space information and thus improve the search of the whole system. In this thesis, a Circle chaotic sequence is used to replace the randomly generated initial population in the EO algorithm, thus improving the diversity of the population [30]. Equation (13) gives the Circle chaotic mapping formula:

$$X_{i+1} = \mathrm{mod}(X_i + 0.2 - (\frac{0.5}{2\pi})\sin(2\pi X_i), 1) \quad i = 1, 2, ..., (n-1) \tag{13}$$

Where mod in the expression is the residual function. Since the Circle chaotic mapping values are between [0, 1], it is necessary to map the Circle chaotic sequence to the search space as shown in equation (14):

$$\vec{C}_i^{initial} = C_{min} + X_i \left(C_{max} - C_{min}\right) \quad i = 1, 2, ..., n \tag{14}$$

2) Levy flight. The exploration phase of the EO algorithm is enhanced by the use of randomized steps based on Levy flights in the EO algorithm. A Levy flight is a random walk, i.e., a sequence of randomized steps, where the randomized steps are determined through the use of a Levy distribution. A Levy distribution is a heavy-tailed distribution, i.e., one with infinite variance. The Levy distribution is shown in Eq. (15):

$$L\left(D, \alpha, \beta\right) = \begin{cases} \sqrt{\dfrac{\beta}{2\pi}} \dfrac{e^{-(\beta/2(D-\alpha))}}{\left(D-\alpha\right)^{3/2}} & 0 < \alpha < D < \infty \\ 0 & else \end{cases} \tag{15}$$

When $D \to \infty$, the Levy distribution is converted to:

$$L(D,\alpha,\beta)=\begin{cases}\sqrt{\dfrac{\beta}{2\pi}}\dfrac{e}{(D-\alpha)^{3/2}} & 0<\alpha<D<\infty \\ 0 & else\end{cases} \tag{16}$$

Eqs. (15) and (16) indicate that the Levy distribution depends on a and β. The values of the random steps using the Levy distribution are as follows:

$$s=\frac{l_1}{l_2^{(Y_\mu)}} \tag{17}$$

Where l_1 and l_2 are random numbers generated using normal distribution given by equation (18).

$$\begin{cases} l_1 = N(0,\sigma^2) \\ l_2 = N(0,1) \end{cases} \tag{18}$$

Where σ^2 is:

$$\sigma^2=\left(\frac{\sin\left(\dfrac{\mu\pi}{2}\right)\Gamma(1+\mu)}{\mu 2^{(\mu-1)}\Gamma\left(\dfrac{1+\mu}{2}\right)}\right)^{1/\mu} \tag{19}$$

Where μ is a parameter that regulates the shape of the Levy distribution, generally let $\mu=1.4$. Γ is a gamma function. Thus, the Levy flight produced in the initial step of the IEO algorithm is given by Eq. (20):

$$L_f(n_d,\mu)=\left[s_1,s_2,...,s_{n_d}\right] \tag{20}$$

Where n_d is the dimension of the particle. In the IEO algorithm, a particle is randomly selected to modify the concentration of each dimension, as shown in equation (21):

$$\vec{C}_i=\vec{C}_i+L_f \tag{21}$$

3) Iterative Cosine Operator (ICO).The EO algorithm works on the basis of an equilibrium pool, which consists of four best particles so far and one particle which is the average of the four best particles. The best four particles help in the exploration capability of the algorithm, while the average particle improves the exploitation capability. Based on the randomly selected particles from the equilibrium pool, the concentration of each particle is updated during each iteration of the EO algorithm. The best particles so far may exist in different regions of the search space or the same region, and if the best particles so far come from different regions, then these elements improve the exploratory power of the algorithm. However, in another case it may fall into local optimality. Therefore, in this paper, we first remove $\vec{C}_{eq(ave)}$ from the

balanced pool and then improve the particles randomly selected from the balanced pool using ICO to improve the convergence to the global optimum as shown in Eq. (22):

$$\vec{C}_{eq} = \begin{cases} \vec{C}_{eq} \times \cos\left(\dfrac{\pi \times Iter}{2 \times Max_i ter}\right) & r_4 < 0.5 \\ \vec{C}_{eq} & r_4 \geq 0.5 \end{cases} \tag{22}$$

Where r_4 is a random number in [0, 1]. Therefore, the particle update formula of the IEO algorithm can be IEO:

$$\vec{C}_i = \begin{cases} \vec{C}_i + L_f & r_3 < 0.2 \\ \vec{C}_{eq} + \left(\vec{C} - \vec{C}_{eq}\right).\vec{F} + \dfrac{\vec{G}}{\vec{\lambda} V}\left(1 - \vec{F}\right) & r_3 \geq 0.2 \end{cases} \tag{23}$$

Where r_3 is a random number in [0, 1] and $\vec{C}_{eg}$ is obtained from equation (18).

The detailed steps of the IEO algorithm are shown below:

Step 1: Initialize the population using Eqs. (14), (15), initialization Max_iter as well as order $Iter = 1$.

Step 2: If $Iter \leq Max_iter$, compute the fitness of individuals and update the equalization pool $\vec{C}_{eq}^{pool}$. Otherwise end the algorithm and return the optimal individual $\vec{C}_{eq(1)}$.

Step 3: Traverse the population of individuals. Let $r_3 = rand(1)$, if $r_3 < 0.2$, update the individual using equation (21). If $r_3 \geq 0.2$, randomly select a particle from the equalization pool $\vec{C}_{eq}^{pool}$ as $\vec{C}_{eq}$ and update $\vec{C}_{eq}$ according to equation (22), compute the index term $\vec{F}$ and generation rate $\vec{G}$, and finally update the individual using equation (23).

Step 4: Order $Iter = Iter + 1$ and jump back to step 2.

4 Machine vision and robot cluster optimization analysis

4.1 Road sign extraction and homing effect analysis

4.1.1 Road Sign Extraction Accuracy Analysis

In order to test the practical application effect of the improved road sign extraction method, this paper takes three positions for each scene in three specific scene spaces for the relocation experiments and calculates the time of matching at each position, the success rate of relocation, and the localization error, respectively. Table 1 shows the comparison between the road sign extraction model in this paper and monocular ORB-SLAM relocation.

The experimental results of ORB-SLAM showed 3 times relocation failure, and this method can match successfully. The average time consumed by ORB-SLAM in single matching is 0.178s, and that of this method is 0.01s. After successful relocation, the robot localization error of this algorithm is lower than 0.05m, and that of the ORB-SLAM algorithm ranges from 0.07 to 0.8 m. The localization errors are lower than 0.05m, and that of the ORB-SLAM algorithm ranges from 0.07 to 0.8m. In summary, the road sign extraction recognition algorithm proposed in this paper has a higher relocation success rate, faster matching speed, and smaller accuracy error after successful matching.

Table 1. Comparison of sign extraction model and monocular orb-slam relocation

	New model matching			ORB-SLAM matching		
	Matching time (s)	Relocation result	Positioning error (m)	Matching time (s)	Relocation result	Positioning error (m)
Scene 1	0.01	Y	0.016	0.178	Y	0.308
	0.01	Y	0.046	0.178	N	0.361
	0.01	Y	0.036	0.178	Y	0.617
Scene 2	0.01	Y	0.033	0.178	Y	0.167
	0.01	Y	0.047	0.178	Y	0.079
	0.01	Y	0.006	0.178	Y	0.315
Scene 3	0.01	Y	0.025	0.178	N	0.599
	0.01	Y	0.009	0.178	N	0.255
	0.01	Y	0.014	0.178	Y	0.802

4.1.2 Cluster robot homing effect analysis

In addition, the average homing error is also calculated in this paper to validate the effectiveness of the proposed swarm robot for visual recognition of road signs and task assignment optimization. Where the metrics are as follows: i.e. (1) Average Moving Time (AMT), i.e., the average moving time for the robot to move from the randomly selected current position to the target position to complete a homing task. (2) Average movement distance (AMD), i.e., the average movement distance for the robot to complete one homing task. (3) Average Moving Steps (AMS), i.e., the average number of moving steps for the robot to complete a homing task. (4) Average angular error (AAE), i.e., the average homing angular error of the robot for completing one homing task. (5) and homing ratio (RR), i.e., the proportion of successful homing times in the total number of homing times.

The actual homing performance statistics of the homing methods are shown in Table 2. The average error of different homing methods was explored by summarizing all the actual experiments for the five indoor environments described in Table 2. The results show that the average robot movement time under the optimization of this algorithm is the shortest in all scenarios. The robot that was optimized by waypoint extraction has an average movement time of 17.25 seconds in the laboratory scenario and 19.83 seconds on the factory floor, which is superior to that of other algorithms. The homing ratio in all scenarios reaches 90.6%, and it is worth mentioning that the robot optimized by this paper's algorithm in a crowded workshop reaches 96%, which is 12% ahead of the second-place VPM algorithm, indicating that the optimized swarm robots in complex scenarios show good homing ability, reflecting that the roadmap extraction algorithm based on the improved Mean Shift algorithm proposed in this paper has good performance.

Table 2. Actual homing performance statics for the homing methods

Scene	Homing method	Experiment frequency	AMT (s)	AMD (m)	AMS	AAE (°)	RR
Lab 1	ALV	50	24.20	7.54	9.22	21.59	0.62
	SLALV		23.49	6.88	10.89	19.80	0.69
	HiSS		25.03	6.24	9.06	25.17	0.61
	MLBH		39.99	5.92	8.96	22.87	0.66
	VPM		24.40	7.44	10.64	17.20	0.74
	New method		**19.48**	**4.82**	**7.29**	**9.20**	**0.87**
Lab 2	ALV	40	21.11	4.36	7.64	18.94	0.72
	SLALV		17.15	4.97	8.13	21.09	0.74
	HiSS		20.09	4.45	7.08	24.01	0.60
	MLBH		31.75	6.19	9.73	15.72	0.68
	VPM		19.74	4.13	7.66	14.50	0.85
	New method		**15.02**	**3.51**	**6.52**	**8.14**	**0.88**
Ordinary factory shop	ALV	35	18.61	4.41	6.99	16.65	0.75
	SLALV		17.50	6.15	7.17	15.84	0.85
	HiSS		16.56	5.14	7.08	19.79	0.69
	MLBH		29.18	5.01	7.53	16.90	0.86
	VPM		16.92	4.57	6.78	12.31	0.83
	New method		**15.31**	**3.52**	**5.33**	**9.40**	**0.94**
Spacious factory shop	ALV	50	30.72	8.16	11.86	21.49	0.74
	SLALV		29.76	7.13	10.65	23.04	0.60
	HiSS		31.01	8.80	10.42	25.60	0.60
	MLBH		47.14	7.72	10.90	13.16	0.81
	VPM		27.88	8.45	11.52	17.44	0.73
	New method		**27.72**	**6.19**	**8.42**	**10.29**	**0.88**
Crowded factory shop	ALV	20	19.71	4.98	6.92	19.25	0.76
	SLALV		19.65	5.15	6.81	21.98	0.74
	HiSS		20.13	6.35	7.34	22.09	0.60
	MLBH		30.31	4.45	7.60	13.59	0.80
	VPM		20.09	3.87	6.24	10.18	0.84
	New method		**16.46**	**3.04**	**5.85**	**7.84**	**0.96**

4.2 Application Analysis of Task Assignment Algorithms - An Example of Medical Dispensing System

Due to the special characteristics of pharmaceuticals, the off-loading times are kept as short as possible. In addition, the operation of the dispensing robot is kept at a constant speed, and the operation time of its startup and unloading are ignored. Based on this, this section proposes an environment model with composite judging criteria that incorporates an adaptation function, and the model functions are shown in Eq. (24) to Eq. (32):

$$F=\min(\sum_{i=1}^{M}\sum_{j=1}^{N}(\mu_{ij}T_{ij}+\lambda_{ij}S_{ij}+\beta_{ij}D_{ij})) \tag{24}$$

$$\sum_{j=1}^{N}(\mu_{ij}+\lambda_{ij}+\beta_{ij})=1 \tag{25}$$

$$M=\{M_1,M_2,\cdots,M_j\} \tag{26}$$

$$M_j=\{m_{j1},m_{j2},\cdots,m_{jk}\} \tag{27}$$

$$T_{ij}=\sum_{Z=m_{j1}}^{m_{jk}}(T_{iZ}^{start}+t_{iZ}) \tag{28}$$

$$S_{ij}=\sum_{Z=m_{j1}}^{m_{jk}}(S_{iZ^pZ}^{start}+s_{iZ^pZ}-s_{iZ^pZ^{Sp}}\times\eta_{iZ^{Sp}}) \tag{29}$$

$$D_{ij}=\sum_{Z=Z_{start}}^{Z_{end}}(\frac{S_{iZ_{start}Z}+S_{iZ}}{v}) \tag{30}$$

$$\beta_{ij}=\begin{cases}a,\text{if the order j contains special drugs}\\0,\text{others}\end{cases} \tag{31}$$

$$\eta_{iZ^{Sp}}=\begin{cases}1,\text{if } Z^{Sp}\text{ is a special drug table}\\0,\text{others}\end{cases} \tag{32}$$

Where, Eq. (24) is the fitness function for dispensing robot scheduling, which is the weighted sum value of all the drug pickup tasks performed by the dispensing robot. In order to ensure efficiency and drug activity, the smaller the fitness function represents, the more powerful the scheduling performance of the dispensing system. Based on this, the scheduling model in this paper requires the minimum value of the fitness function to be the optimal value for the solution of this paper.

4.2.1 Accuracy and Stability Analysis of Tasking Algorithms

The scheduling algorithm data can be found in Table 3. This paper's cluster robot task allocation algorithm is superior to both the HS algorithm and the IGHS algorithm in terms of scheduling task calculation speed and accuracy. Among them, when the task volume is 30, 300, and 1000, the optimization rate of the total fitness average of this paper's algorithm relative to HS and IGHS algorithms reaches 17.35%, 4.26%, 2.24% and 15.76%, 3.88% and 1.51%, respectively, and the accuracy improvement of single optimal scheduling fitness data reaches 13.34%, 5.39%, 2.81% and 10.95%, 4.09%, and 1.77%, respectively. On the whole, the scheduling effect of the cluster robot task allocation algorithm proposed in this paper is more stable, the scheduling results are highly accurate, and it can complete scheduling tasks of different complexity, which is conducive to the medical automatic dispensing system to quickly, accurately, and efficiently complete the tasks of taking medication, and to promote the overall performance of the system.

Table 3. Scheduling algorithm data

Task quantity		HS	Our method	IGHS
30	Mean	**72.46**	**59.89**	**71.09**
	Best	63.71	55.21	62.00
	Worst	91.45	69.07	79.56
	Std	2.80	1.37	4.85
300	Mean	**807.49**	**773.08**	**804.24**
	Best	800.96	757.80	790.10
	Worst	818.79	782.84	816.31
	Std	5.07	1.80	8.50
1000	Mean	**2766.59**	**2704.67**	**2746.09**
	Best	2751.21	2673.91	2722.16
	Worst	2814.61	2718.77	2782.45
	Std	14.55	12.05	18.58

The stability analysis of the task scheduling algorithm can be seen in Figure 3. This paper's task scheduling algorithm has a more stable distribution of results when dealing with tasks of different complexity. When the amount of tasks is 30, 300, and 1000, the standard deviation of this paper's algorithm is 1.37, 1.8, and 12.05, respectively, which are all significantly smaller than the other two algorithms, as reflected in the box-and-line diagrams, i.e., the data distribution is more centralized.

In summary, the ability of this paper's algorithm to search for optimal solutions is much higher than that of the comparison algorithms, and it can handle all kinds of tasks in a timely and effective manner with superior results in the process of dispensing robot cluster scheduling. In terms of data distribution and mathematical statistics, this paper's algorithm is more centralized for various types of scheduling tasks, and it is stable and highly reliable.

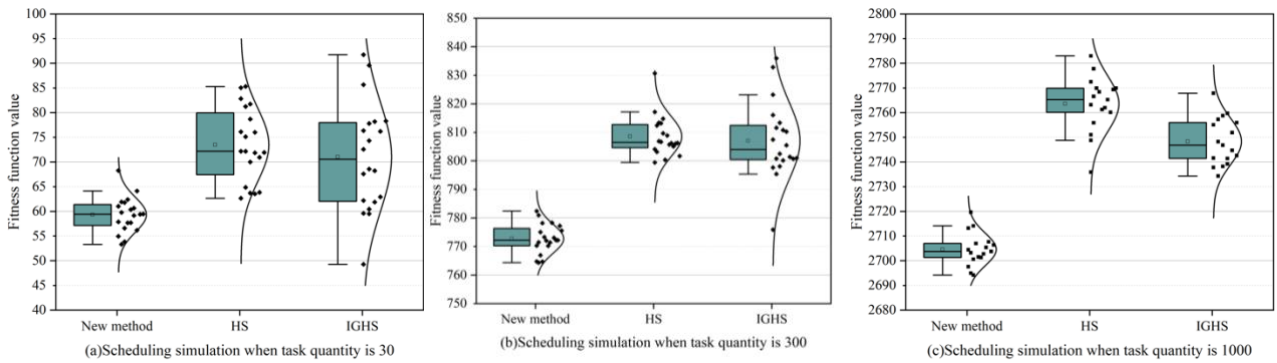


Figure 3. Analysis of the stability of the task distribution algorithm

4.2.2 Efficiency analysis of tasking algorithms

In this paper, a total of three mobile robot clusters, referred to as AGVs, are set to depart at 1s intervals in order to avoid the conflict of multiple AGVs departing from the starting point at the same time. In addition, *AGV is the mobile robot responsible for the remaining drug dispensing tasks in the order of scheduling drug orders. According to the difference of the earliest mobile robot that completes the order task, the scheduling mobile robot responsible for the remaining drug dispensing task in the new algorithm is AGV1, and the scheduling mobile robot responsible for the remaining drug dispensing

task in the HS algorithm is AGV3. Meanwhile, the scheduling mobile robot responsible for the remaining drug dispensing task *AGV needs to complete the initial order according to the initial order of the mobile robot before carrying out the remaining drug dispensing work. Before performing the remaining drug dispensing work, the scheduling mobile robot *AGV needs to move to the starting point of the mobile robot according to the final position of the mobile robot that has completed the initial order (scheduling target point) and the time used is credited to the total scheduling time.

The mobile robot scheduling operation data table is shown in Table 4. For the task allocation algorithm proposed in this paper, the total time for the mobile robot to complete all order tasks from t_0 =0s t_{Total} =125.25s, which is a 9.56% improvement in scheduling task completion efficiency compared to the total time for the mobile robot to complete all order tasks in the comparison algorithm HS t_{Total} =138.49s.

Table 4. Mobile robot scheduling running data tables

	Index	AGV1	AGV2	AGV3	*AGV
New method	t1	41.78	23.78	28.52	105.01
	t2	49.62	38.74	40.15	
	t3	59.67	39.72	53.89	
	Total	90.14	66.91	73.50	**125.25**
HS	t1	34.80	29.84	22.57	100.52
	t2	42.37	36.64	30.03	
	t3	60.57	52.92	30.95	
	Total	91.00	78.21	58.70	**138.49**

5 Conclusion

Two aspects of industrial swarm robot control optimization are discussed in this paper: the extraction and matching of binocular image feature points and the task allocation algorithm of the swarm robot. The roadmap extraction accuracy analysis shows that the method in this paper can successfully match all roadmaps in the experiments; the average error of localization is 0.025 m in all the experimental scenarios, and the extreme value of the error is less than 0.05 m. The homing effect analysis shows that in the laboratory scenarios, the robot optimized by the new algorithm has an average movement time of 17.25 s. The average movement time in a factory floor is 19.83 s, which is better than the performance of other algorithms. The average movement time and homing ratio in all scenarios is 90%. This algorithm is better than other algorithms, and the homing ratio is 90.6% in all scenarios.

The performance of the task assignment algorithm is also examined, and it is found that the algorithm in this paper optimizes 17.35%, 4.26%, 2.24%, and 15.76%, 3.88%, 1.51% of the total fitness averages with respect to the HS and IGHS algorithms, and the algorithms have small standard deviations of 1.37, 1.8, and 12.05 under the three kinds of task volumes, respectively. In the medical dispensing experiment, the total time for the mobile robot to complete all the order tasks is 125.25s, which improves the scheduling task completion efficiency by 9.56% compared with the 138.49s of the comparison algorithm.

The roadmap extraction algorithm and cluster robot task allocation algorithm proposed in this paper both perform well and contribute to improving the collaboration efficiency of cluster industrial robots.

Funding:

1) Supported by Surface Project of Natural Science Research in Jiangsu Higher Education Institutions (22KJD510012).

2) Jiangsu Provincial Department of Education University Youth and Blue Project Program Industrial Robotics Technology Teaching Team (Su Teacher's letter No. [2022] 29).

3) Yangzhou Science and Technology Program Municipal School Cooperation Project Funding (YZ2022181).

References

[1] Ali, M. H., Aizat, K., Yerkhan, K., Zhandos, T., & Anuar, O. (2018). Vision-based robot manipulator for industrial applications. Procedia computer science, 133, 205-212.

[2] Ren, Z., Fang, F., Yan, N., & Wu, Y. (2022). State of the art in defect detection based on machine vision. International Journal of Precision Engineering and Manufacturing-Green Technology, 9(2), 661-691.

[3] Yen, V. T., Nan, W. Y., & Van Cuong, P. (2019). Recurrent fuzzy wavelet neural networks based on robust adaptive sliding mode control for industrial robot manipulators. Neural Computing and Applications, 31(11), 6945-6958.

[4] Steger, C., Ulrich, M., & Wiedemann, C. (2018). Machine vision algorithms and applications. John Wiley & Sons.

[5] Urrea, C., & Pascal, J. (2018). Design, simulation, comparison and evaluation of parameter identification methods for an industrial robot. Computers & electrical engineering, 67, 791-806.

[6] Lins, R. G., de Araujo, P. R. M., & Corazzim, M. (2020). In-process machine vision monitoring of tool wear for Cyber-Physical Production Systems. Robotics and computer-integrated manufacturing, 61, 101859.

[7] Brito, T., Queiroz, J., Piardi, L., Fernandes, L. A., Lima, J., & Leitão, P. (2020). A machine learning approach for collaborative robot smart manufacturing inspection for quality control systems. Procedia Manufacturing, 51, 11-18.

[8] Fernández-Robles, L., Azzopardi, G., Alegre, E., & Petkov, N. (2017). Machine-vision-based identification of broken inserts in edge profile milling heads. Robotics and Computer-Integrated Manufacturing, 44, 276-283.

[9] Yao, B., Zhou, Z., Wang, L., Xu, W., Liu, Q., & Liu, A. (2018). Sensorless and adaptive admittance control of industrial robot in physical human− robot interaction. Robotics and Computer-Integrated Manufacturing, 51, 158-168.

[10] Zhou, L., Zhang, L., & Konz, N. (2022). Computer vision techniques in manufacturing. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 53(1), 105-117.

[11] Wang, T., Chen, B., Zhang, Z., Li, H., & Zhang, M. (2022). Applications of machine vision in agricultural robot navigation: A review. Computers and Electronics in Agriculture, 198, 107085.

[12] Quarta, D., Pogliani, M., Polino, M., Maggi, F., Zanchettin, A. M., & Zanero, S. (2017, May). An experimental security analysis of an industrial robot controller. In 2017 IEEE Symposium on Security and Privacy (SP) (pp. 268-286). IEEE.

[13] Javaid, M., Haleem, A., Singh, R. P., Rab, S., & Suman, R. (2022). Exploring impact and features of machine vision for progressive industry 4.0 culture. Sensors International, 3, 100132.

[14] Alonso, V., Dacal-Nieto, A., Barreto, L., Amaral, A., & Rivero, E. (2019). Industry 4.0 implications in machine vision metrology: an overview. Procedia manufacturing, 41, 359-366.

[15] Khang, A., Hajimahmud, V. A., Ali, R. N., Hahanov, V., & Avramovic, Z. (2024). Role of Machine Vision in Manufacturing and Industrial Revolution 4.0. In Machine Vision and Industrial Robotics in Manufacturing (pp. 1-13). CRC Press.

[16] Moru, D. K., & Borro, D. (2020). A machine vision algorithm for quality control inspection of gears. The International Journal of Advanced Manufacturing Technology, 106(1), 105-123.

[17] Penumuru, D. P., Muthuswamy, S., & Karumbu, P. (2020). Identification and classification of materials using machine vision and machine learning in the context of industry 4.0. Journal of Intelligent Manufacturing, 31(5), 1229-1241.

[18] Frank, D., Chhor, J., & Schmitt, R. (2017, December). Stereo-vision for autonomous industrial inspection robots. In 2017 IEEE International Conference on Robotics and Biomimetics (ROBIO) (pp. 2555-2561). IEEE.

[19] Benbarrad, T., Salhaoui, M., Kenitar, S. B., & Arioua, M. (2021). Intelligent machine vision model for defective product inspection based on machine learning. Journal of Sensor and Actuator Networks, 10(1), 7.

[20] Arents, J., & Greitans, M. (2022). Smart industrial robot control trends, challenges and opportunities within manufacturing. Applied Sciences, 12(2), 937.

[21] Evjemo, L. D., Gjerstad, T., Grøtli, E. I., & Sziebig, G. (2020). Trends in smart manufacturing: Role of humans and industrial robots in smart factories. Current Robotics Reports, 1, 35-41.

[22] Vick, A., & Krueger, J. (2018, June). Using OPC UA for distributed industrial robot control. In ISR 2018; 50th International Symposium on Robotics (pp. 1-6). VDE.

[23] Wang, Y. (2020). Robot algorithm based on neural network and intelligent predictive control. Journal of Ambient Intelligence and Humanized Computing, 11(12), 6155-6166.

[24] Qian, J., Zi, B., Wang, D., Ma, Y., & Zhang, D. (2017). The design and development of an omni-directional mobile robot oriented to an intelligent manufacturing system. Sensors, 17(9), 2073.

[25] Qiangxian Huang,Tao Xiang,Zhihao Zhao,Kui Wu,Hongli Li,Rongjun Cheng... & Zhenying Cheng. (2024). Directional region-based feature point matching algorithm based on SURF. Journal of the Optical Society of America. A, Optics, image science, and vision(2),157-164.

[26] Han Luyang,Boese Markus,Gamm Bjoern & Tordoff Benjamin. (2021). A new beam alignment method in SEM based on parallax principle. Microscopy and Microanalysis(S1),1612-1613.

[27] Cui Yongbin. (2024). Application of cultural elements of dunhuang murals in landscape design based on mean shift algorithm extraction. Journal of Computational Methods in Sciences and Engineering(1),473-487.

[28] Yang Jinfeng,Que Huakun,Liu Wenjia & Xiao Jiang. (2024). A Monitoring Model for Abnormal Electricity Consumption Based on K-Means++ Clustering and Improved K-Nearest Neighbor Algorithm. Smart Grids and Sustainable Energy(2).

[29] N. Hanuman Reddy,Lathigara Amit,Aluvalu Rajanikanth & V. Uma Maheswari. (2024). Clustering based EO with MRF technique for effective load balancing in cloud computing. International Journal of Pervasive Computing and Communications(1),168-192.

[30] Wang Wentao & Tian Jun. (2022). An Improved Nonlinear Tuna Swarm Optimization Algorithm Based on Circle Chaos Map and Levy Flight Operator. Electronics(22),3678-3678.

About the Authors

Linyang Guo (1983-), male, from Jinan, Jilin province, graduate student, associate professor, senior engineer, mainly engaged in machine vision, automatic control and optoelectronic technology research work.

Runxian Yang (1979-), female, from Shaanxi Heyang province, optoelectronic technology and other aspects of the research work.